

Lecture 11

Factor Analysis and Structural Equation Modelling

FACTOR ANALYSIS

Let $Y_i, i = 1, 2, \dots, n$ be independent $N_p(\mu, \Sigma)$ variables (think of the Y_i 's as a results of a battery of p tests applied to the i -th individual). Fundamental assumption in factor analysis:

$$Y_i = \Lambda f_i + e_i \quad (1)$$

– $\Lambda \in \mathcal{M}_{p,k}$ – factor loading matrix (full rank);

– $f_i \in R^k (k < p)$ – factor variable. The components of f_i are thought to be the (latent) factors. Usually f_i are taken to be independent $N(\alpha, I_k)$ (i.e. "orthogonal") but also "oblique" factors are considered sometimes with a covariance matrix $\neq I_k$.

– The vectors e_i – independent $N(\theta, \Sigma_e)$ with Σ_e – **diagonal**, i.e. $\Sigma_e = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_p^2)$.

Also, the e 's are independent from the f 's.

Hence

$$\mu = \Lambda\alpha + \theta, \quad \Sigma = \Lambda\Lambda' + \Sigma_e$$

or, component wise:

$$\text{Var}(Y_{ir}) = \sum_{j=1}^k \lambda_{rj}^2 + \sigma_r^2 = \text{communality} + \text{uniqueness} .$$

$$\text{Cov}(Y_{ir}, Y_{is}) = \sum_{j=1}^k \lambda_{rj} \cdot \lambda_{sj}$$

The fundamental idea of factor analysis is to describe the **covariance relationships** among **many** variables (p large) in terms of view (k "small") underlying, not observable (latent) random quantities (the **factors**). The model is motivated by the following argument: suppose variables can be grouped by their correlations. That is , all variables in a particular group are highly correlated among themselves but have relatively small correlations with variables in a different group. It is then quite reasonable to assume that each group of variables represents a single underlying construct (**factor**) that is "responsible" for the observed correlations.

Important Notes:

– The model (1) is similar to a linear regression model but the key differences are that f_i are **random and are not observable**.

– If we knew the Λ (or have found estimates of them), then using properties of orthogonal projections on the linear space spanned by the columns of Λ , we would get:

$$\hat{\alpha} = (\Lambda' \Lambda)^{-1} \Lambda' \bar{Y}; \hat{\theta} = \bar{Y} - \Lambda \hat{\alpha}.$$

Because of the above observation, we can consider only μ, Λ , and $\sigma_i^2, i = 1, 2, \dots, p$ as unknown parameters when parameterising the factor analysis model. Note also that primary interest in factor analysis is focused on estimating Λ .

Note also that **there is a fundamental indeterminacy** in this model even when we require that $\text{Var}(\mathbf{f}) = I_k$ since, if $P \in \mathcal{M}_{k,k}$ is any orthogonal matrix then obviously

$$\Lambda\Lambda' = \Lambda P(\Lambda P)'; \Lambda f_i = (\Lambda P)(P' f_i).$$

Hence replacing Λ by ΛP and f_i by $P' f_i$ leads to the same equations.

I. ML Estimation. The likelihood function for the n observations $\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_n \in \mathbf{R}^p$ is:

$$\begin{aligned} L(\mathbf{Y}; \mu, \Lambda, \sigma_1^2, \sigma_2^2, \dots, \sigma_p^2) &= (2\pi)^{-np/2} |\Sigma|^{-n/2} \exp \left[-\frac{1}{2} \sum_{i=1}^n (\mathbf{Y}_i - \mu)' \Sigma^{-1} (\mathbf{Y}_i - \mu) \right] = \\ &= (2\pi)^{-np/2} |\Sigma|^{-n/2} \cdot \exp \left[-\frac{n}{2} (\text{tr} \Sigma^{-1} S + (\bar{Y} - \mu)' \Sigma^{-1} (\bar{Y} - \mu)) \right] \end{aligned}$$

with $S = \frac{1}{n} \sum_{i=1}^n (\mathbf{Y}_i - \bar{Y})(\mathbf{Y}_i - \bar{Y})'$

Taking $\ln L$, we get:

$$\begin{aligned} \ln L(\mathbf{Y}; \mu, \Lambda, \sigma_1^2, \sigma_2^2, \dots, \sigma_p^2) &= -\frac{np}{2} \ln(2\pi) - \frac{n}{2} \ln(|\Sigma|) \\ &\quad - \frac{n}{2} [\text{tr}(\Sigma^{-1} S) + (\bar{Y} - \mu)' \Sigma^{-1} (\bar{Y} - \mu)] \end{aligned}$$

After differentiating w.t. μ :

$$\frac{\partial \ln L}{\partial \mu} = n \Sigma^{-1} (\bar{Y} - \mu) = 0 \longrightarrow \hat{\mu} = \bar{Y}.$$

It remains to estimate Λ, Σ_e by minimizing:

$$Q = \frac{1}{2} \ln |\Lambda\Lambda' + \Sigma_e| + \frac{1}{2} \text{tr}(\Lambda\Lambda' + \Sigma_e)^{-1} S$$

To implement the minimization of Q we use the following rules for matrix differentiation:

$$\frac{\partial}{\partial \Lambda} \ln |\Lambda\Lambda' + \Sigma_e| = 2(\Lambda\Lambda' + \Sigma_e)^{-1} \Lambda \quad (2)$$

$$\frac{\partial}{\partial A} \text{tr}(A^{-1} B) = -(A^{-1} B A^{-1})' \quad (3)$$

Applying (3) and the chain rule we get:

$$\frac{\partial}{\partial \Lambda} \text{tr}[(\Lambda\Lambda' + \Sigma_e)^{-1} S] = -2(\Lambda\Lambda' + \Sigma_e)^{-1} S (\Lambda\Lambda' + \Sigma_e)^{-1} \Lambda$$

Hence after substitution:

$$\begin{aligned} \frac{\partial}{\partial \Lambda} Q &= (\Lambda\Lambda' + \Sigma_e)^{-1} \Lambda - (\Lambda\Lambda' + \Sigma_e)^{-1} S (\Lambda\Lambda' + \Sigma_e)^{-1} \Lambda = \\ &= (\Lambda\Lambda' + \Sigma_e)^{-1} [\Lambda\Lambda' + \Sigma_e - S] (\Lambda\Lambda' + \Sigma_e)^{-1} \Lambda = 0. \end{aligned} \quad (4)$$

A direct check shows that

$$(\Lambda\Lambda' + \Sigma_e)^{-1} = \Sigma_e^{-1} - \Sigma_e^{-1}\Lambda(I + \Lambda'\Sigma_e^{-1}\Lambda)^{-1}\Lambda'\Sigma_e^{-1} \quad (5)$$

Hence from (4) and (5) we get

$$[\Lambda\Lambda' + \Sigma_e - S]\Sigma_e^{-1}\Lambda\{I - (I + \Lambda'\Sigma_e^{-1}\Lambda)^{-1}\Lambda'\Sigma_e^{-1}\Lambda\} = 0 \quad (6)$$

Since the rank of the matrix in the curly brackets in (6) is full we get:

$$[\Lambda\Lambda' + \Sigma_e - S]\Sigma_e^{-1}\Lambda = 0$$

or equivalently:

$$S\Sigma_e^{-1}\Lambda = \Lambda(I + \Lambda'\Sigma_e^{-1}\Lambda)$$

The latter can also be written as:

$$(\Sigma_e^{-1/2}S\Sigma_e^{-1/2})\Sigma_e^{-1/2}\Lambda = \Sigma_e^{-1/2}\Lambda(I + \Lambda'\Sigma_e^{-1}\Lambda) \quad (6).$$

To find a particular solution, **we require $\Lambda'\Sigma_e^{-1}\Lambda$ to be diagonal**. Then (6) implies that the matrix $\Sigma_e^{-1/2}\Lambda$ has as its columns k eigenvectors that correspond to the k eigenvalues of $\Sigma_e^{-1/2}S\Sigma_e^{-1/2}$. More subtle analysis shows that to obtain the minimum value of Q these have to be the eigenvectors that correspond to the **largest** eigenvalues of $\Sigma_e^{-1/2}S\Sigma_e^{-1/2}$.

Based on this fact, the following iterative solution (due to Lawley) has been proposed that can be described algorithmically as follows:

With an initial guess $\tilde{\Sigma}_e \longrightarrow$ calculate $\tilde{\Sigma}_e^{-1/2}\tilde{\Lambda}$ by using the eigenvectors of the k largest eigenvalues of $\tilde{\Sigma}_e^{-1/2}S\tilde{\Sigma}_e^{-1/2}$. Then from $\tilde{\Sigma}_e^{-1/2}\tilde{\Lambda} \longrightarrow$ get a (first iteration) value for $\tilde{\Lambda}$. With this value of $\tilde{\Lambda}$ we can calculate the value of

$$\tilde{Q}(\tilde{\Sigma}_e) = \frac{1}{2} \ln |\tilde{\Lambda}\tilde{\Lambda}' + \tilde{\Sigma}_e| + \frac{1}{2} \text{tr}(\tilde{\Lambda}\tilde{\Lambda}' + \tilde{\Sigma}_e)^{-1}S \text{ (which is the value of the functional).}$$

This functional only depends on the p non-zero values of $\tilde{\Sigma}_e$ and there are several powerful numerical procedures to find its minimum. If it is achieved at Σ_e^* , then calculate $(\Sigma_e^*)^{-1/2}\Lambda^*$ by using the eigenvectors of the k largest eigenvalues of $(\Sigma_e^*)^{-1/2}S(\Sigma_e^*)^{-1/2}$. Then from $(\Sigma_e^*)^{-1/2}\Lambda^* \longrightarrow$ get the (last iterated) value for Λ^* .

II. Hypothesis testing under multivariate normality assumption.

The most interesting hypothesis is $H_0 : k \text{ factors against } H_1 : \neq k \text{ factors}$.

$$\ln L_\Omega = -\frac{np}{2} \ln(2\pi) - \frac{n}{2} \ln |S| - \frac{np}{2} ; \ln L_\omega = -\frac{np}{2} \ln(2\pi) - \frac{n}{2} \ln |\hat{\Sigma}| - \frac{n}{2} \text{tr}(\hat{\Sigma}^{-1}S) \text{ (where } \hat{\Sigma} = \hat{\Lambda}\hat{\Lambda}' + \hat{\Sigma}_e).$$

Hence $-2 \ln \frac{L_\omega}{L_\Omega} = n[\ln |\hat{\Sigma}| - \ln |S| + \text{tr}(\hat{\Sigma}^{-1}S) - p]$. The asymptotic distribution of this statistic is χ^2 with $df = \frac{p(p+1)}{2} - [pk + p - \frac{k(k-1)}{2}] = \frac{1}{2}[(p-k)^2 - p - k]$ WHY!

III. Varimax method of rotating the factors

If Λ_0 is the estimated factor loading matrix obtained by the ML method, we know that $\Lambda = \Lambda_0 P$ with any orthogonal $P \in \mathcal{M}_{k,k}$ can be used instead. How to choose a particular P such that Λ has some desirable properties?

Let $d_r = \sum_{i=1}^p \lambda_{ir}^2$, then the **varimax method of rotating the factors** consists in choosing P to maximize

$$S = \sum_{r=1}^k \left\{ \sum_{i=1}^p (\lambda_{ir}^2 - \frac{d_r}{p})^2 \right\} = \sum_{r=1}^k \left\{ \sum_{i=1}^p \lambda_{ir}^4 - \frac{(\sum_{i=1}^p \lambda_{ir}^2)^2}{p} \right\}$$

This corresponds to the wish to make, for each column of factor loadings, some of the coordinates to be "very large" and the rest to be "very small" (in absolute value). Iterative solution to the above rotation problem exists.

Note: Rotation of factor loadings is **particularly recommended** for loadings obtained by ML method since the initial values of Λ_0 are constrained to satisfy the condition $\Lambda_0' \Sigma_e^{-1} \Lambda_0$ -**diagonal**. This is convenient for computational purposes but may not lead to easily interpretable factors.

IV. Relation to Principal Component Analysis

There are different ways in which you can relate Factor analysis to Principal Component analysis. We will discuss two of them here.

1) The principal component solution of the factor model. Starting with the matrix

$$S = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})(Y_i - \bar{Y})'$$

we can write down its spectral decomposition by using **all** of its p eigenvalues and eigenvectors. In such a way we would derive a perfect reconstruction of S but since it has been achieved by using p factors, it does not deliver any dimension reduction and is useless. We would prefer to employ a smaller number k of eigenvalues and eigenvectors of S and to get only an approximate reconstruction of S

$$S \approx \sum_{i=1}^k \tau_i \vec{a}_i \vec{a}_i' = \Lambda \Lambda'$$

whereby τ_i are the characteristic roots of S , taking the k biggest ones (w.o.l.g. $\tau_1, \tau_2, \dots, \tau_k$) and $\vec{a}_i$ being their corresponding eigenvectors. Since the understanding is that (if k is the right number of factors) all communalities have been taken into account then $s_{ii} - \sum_{j=1}^k \lambda_{ij}^2$ would be the estimators of the uniquenesses. This approach shows the k factors have been extracted from S in the same way like the principal components are calculated. The method is called **the principal component solution of the factor model**.

2) The Principal Factor Solution. This is yet another method that uses similar ideas from principal components analysis. It is similar to the principal component solution but the factor extraction is performed not directly on S . To describe it, let us assume for a moment that the uniquenesses are known (or can be estimated reasonably well) and we can decompose

$$S = S_r + \Sigma_e$$

whereby the number k of factors is known and Σ_e is the diagonal matrix containing the uniquenesses. Then the factor analysis model states that (an estimate of) Λ should satisfy:

$$S_r = S - \Sigma_e = \Lambda\Lambda'$$

Hence Λ estimate can be found by performing principal component analysis on S_r :

If $S_r = \sum_{i=1}^p t_i \vec{\mathbf{b}}_i \vec{\mathbf{b}}_i'$, t_i being the characteristic roots of S_r , take the k biggest ones (w.o.l.g. $t_1, t_2, \dots, t_k$). Denote

$$\mathbf{B}_{p \times p} = (\mathbf{B}_{1(p \times k)} | \mathbf{B}_{2(p \times (p-k))}); \Delta = \text{diag}(t_1, t_2, \dots, t_k). \text{ Then}$$

Then $\hat{\Lambda} = \mathbf{B}_1 \cdot \Delta^{1/2}$. Can do it also iteratively!

Problems:

i) no reliable estimate of Σ_e available. (The most commonly used one in the case where S is the **correlation** matrix R is $\sigma_{ei}^2 = 1/r^{ii}$ where r^{ii} is the i th diagonal element of R^{-1} .)

ii) how to select k .

Note: The methods in Section IV are not efficient as compared to the ML method and in general, the ML method is the preferred one. However, for the ML method one has to assume normality and the alternative approaches described here are used in cases where multivariate normality is in a serious doubt. Most often in practice the choice of k is done by combining subject matter knowledge, "reasonableness" of results and by looking at proportion variance explained.

Example. Data about five socioeconomic variables for 12 census data in the Los Angeles area. The five variables represent total population, median school years, total unemployment, miscellaneous professional services, and median house value. The data is in the file 'factor.dat'. Perform a factor analysis on the above data set. Use ML method and varimax rotation.

SAS code:

```
data soecon; infile 'factor.dat';
input pop school employ services house;
proc factor data=soecon method=ml heywood n = 2 rotate=varimax
reorder plot; run;
```

Some remarks w.r. to the **factor procedure** in SAS:

- if you want to extract different numbers of factors (the code gives you an example how to extract $n = 2$ factors), you should run the procedure once for each number of factors;

- the communalities need a preliminary estimates. If one considers the **correlation matrix** instead of Σ , then the communalities can be estimated by the squared multiple correlations of each of the variables with the rest (these communality-estimates are used to get preliminary estimates of the uniquenesses to start the iteration process). If in the **iteration process** it happens that a communality estimate exceeds 1-the case is referred to as an **ultra-Heywood** case and the Heywood option sets such communality to one thus allowing iterations to continue

- the **scree** option can be used to produce a plot of the eigenvalues Σ that is helpful in deciding how many factors to use

- besides **method=ml** you can use **method=principal** ;

- with the ML method option, the Akaike's Information criterion (and Schwarz's Bayesian Criterion) are included. These can be used to estimate the "best" number of parameters to include in a model (in case more than one model is acceptable). The number of factors that yields the smallest AIC- value is considered "best".

- Try to run the above model with $n = 3$ factors $\rightarrow$ the message "WARNING: Too many factors for a unique solution" appears. This is not surprising as the number of parameters in the model will exceed the number of elements in $\Sigma (\frac{1}{2}[(p-k)^2 - p - k] = -2)$. In this example you can run the procedure for $n = 1$ and for $n = 2$ only (do it!) and you will see that $n = 2$ gives the adequate representation.

STRUCTURAL EQUATION MODELLING

Factor analysis (FA) is only one example of a new approach to data analysis which is **not based on the individual observations**. We were not able to use the regression approach since the input factors were **latent** (not observable). There were too many unknowns. We went to analyse the covariance matrix Σ (and its estimator S) which involved the actual parameters of interest- σ_i^2 and Λ . That is, we switched **from the level of individual observations** to analyze covariance-matrices instead.

There are a **series** of methods which are based on analysis of **covariances** rather than individual cases. Instead of minimizing functions of observed and predicted **individual values**, we minimize the differences between **sample covariances and covariances predicted by the model**.

The fundamental hypothesis in these analyses is

$$H_0 : \Sigma = \Sigma(\theta) \quad \text{against} \quad H_1 : \Sigma \neq \Sigma(\theta)$$

Here Σ has $p(p+1)/2$ unknown elements (estimated by S) **but these are assumed to be reproducible by just $k = \dim(\theta) < p(p+1)/2$ parameters**.

Note that more generally we could consider fitting **means and covariances, or means and covariances and higher moments** to a given structure.

Regression analysis with random inputs, simultaneous equations systems, confirmatory factor analysis, canonical correlations, (M)ANOVA can be considered special cases.

Structural equation modeling is an important statistical tool in economics and behavioural sciences. Structural equations express relationships among several variables that can be either directly observed variables (manifest variables) or unobserved hypothetical variables (latent variables). In **structural models**, as opposed to **functional models**, all variables are taken to be **random** rather than having fixed levels. In addition, for maximum likelihood estimation and generalized least - squares estimation (see below), the random variables are assumed to have an approximately multivariate normal distribution. Hence you are advised to remove outliers and consider transformations to normality before using e.g. the SAS procedure CALIS that is designed to analyze structural equation models.

General form of the model

$$(1) \quad \eta = B\eta + \Gamma\xi + \zeta$$

where $B \in m_{m \times m}$, $\Gamma \in m_{m \times n'}$, $\eta \in \mathbf{R}^m$ -vector of output latent variables:

$\xi \in \mathbf{R}^{n'}$ -vector of input latent variables;

B, Γ -coefficient matrices

Note: $(I - B)$ is assumed to be non-singular.

$\zeta \in \mathbf{R}^m$ - disturbance vector with $E\zeta = 0$.

To this **modelling equation (1)** we attached 2 **measurement equations**:

$$(2) \quad Y = \Lambda_y \eta + \epsilon;$$

$$(3) \quad X = \Lambda_X \xi + \delta ; Y \in \mathbf{R}^p, X \in \mathbf{R}^q; \Lambda_Y \in m_{p \times m}, \Lambda_X \in m_{q \times n'}$$

$\epsilon \in \mathbf{R}^p$, $\delta \in \mathbf{R}^q$ being zero-mean measurement errors. These errors are assumed to be uncorrelated with ξ and ζ and with each other. The above quite general model (1)-(2)-(3) is called **Keesling-Wiley-Jöreskog** model. Its interpretation is that the input and output latent variables ξ and η are connected by a system of linear equations (the structural model (1)) with coefficient matrices B and Γ and an error vector ζ . The random vectors Y and X represent the observable vectors (measurements).

The **implied covariance matrix for this model can be obtained**.

Let

$\text{Var}(\xi) = \Phi$; $\text{Var}(\zeta) = \Psi$; $\text{Var}(\epsilon) = \Theta_\epsilon$; $\text{Var}(\delta) = \Theta_\delta$, then:

$$(4) \quad \Sigma = \Sigma(\theta) = \begin{pmatrix} \Sigma_{YY}(\theta) & \Sigma_{YX}(\theta) \\ \Sigma_{XY}(\theta) & \Sigma_{XX}(\theta) \end{pmatrix}$$

$$= \begin{pmatrix} \Lambda_Y(I - B)^{-1}(\Gamma\Phi\Gamma' + \Psi)[(I - B)^{-1}]'\Lambda_Y' + \Theta_\epsilon & \Lambda_Y(I - B)^{-1}\Gamma\Phi\Lambda_X' \\ \Lambda_X\Phi\Gamma'[(I - B)^{-1}]'\Lambda_Y' & \Lambda_X\Phi\Lambda_X' + \Theta_\delta \end{pmatrix}$$

I Estimation:

MLE method (under normality assumption)

Since the “data” is the estimated covariance matrix $S = \frac{1}{n-1} \sum_{i=1}^n \left\{ \begin{pmatrix} Y_i - \bar{Y} \\ X_i - \bar{X} \end{pmatrix} \cdot \begin{pmatrix} Y_i - \bar{Y} \\ X_i - \bar{X} \end{pmatrix}' \right\}$

and since it is known that $(n-1)S \sim W_{p+q}(n-1, \Sigma, 0)$, we can utilize the form of the Wishart density to derive that

$$\ln L(S, \Sigma(\theta)) = \text{constant} - \frac{n-1}{2} \{ \ln |\Sigma(\theta)| + \text{tr}[S\Sigma^{-1}(\theta)] \}$$

This is the function that has to be maximized. Hence, to find MLE, we minimize:

$$(5) \quad F_{ML}(\theta) = \ln |\Sigma(\theta)| + \text{tr}[S\Sigma^{-1}(\theta)] - \ln |S| - (p+q)$$

The function (5) has the advantage that F_{ML} would be zero for the ‘saturated model’ (with $\hat{\Sigma} = S$), i.e.. a perfect fit is indicated by zero (and any non-perfect fit gives rise to > 0 value of F_{ML}).

II. Model evaluation:

Under normality, model adequacy is mostly tested by an asymptotic χ^2 -test. Under $H_0 : \Sigma = \Sigma(\theta)$ versus $H_1 : \Sigma \neq \Sigma(\theta)$, the statistic to be used is $T = (n-1)F_{ML}(\hat{\theta}_{ML})$ and under H_0 , its asymptotic distribution is χ^2 with $df = \frac{(p+q)(p+q+1)}{2} - \dim(\theta)$

Reason:

$\ln L_\omega = \ln L(S, \hat{\Sigma}_{MLE}) = \ln L(S, \Sigma(\hat{\theta}_{ML})) = -\frac{n-1}{2} \{ \ln |\hat{\Sigma}_{MLE}| + \text{tr}[S\hat{\Sigma}_{MLE}^{-1}] \} + \text{constant};$

$$\ln L_\Omega = \ln L(S, S) = -\frac{n-1}{2} \{ \ln |S| + (p+q) \} + \text{constant}$$

$$\begin{aligned} \text{Hence } -2 \ln \frac{L_\omega}{L_\Omega} &= (n-1) \{ \ln |\hat{\Sigma}_{MLE}| + \text{tr}(S\hat{\Sigma}_{MLE}^{-1}) - \ln |S| - (p+q) \} = \\ &= (n-1)F_{ML}(\hat{\theta}_{ML}) \end{aligned}$$

III. SOME PARTICULAR SEM

From the general model 1)-2)-3), we can obtain following particular models:

A) $\Lambda_Y = I_m$, $\Lambda_X = I_{n'}$; $p = m$; $q = n'$; $\Theta_\epsilon = 0$; $\Theta_\delta = 0 \longrightarrow Y = BY + \Gamma X + \zeta$ (the classical econometric model).

B) $\Lambda_Y = I_p$, $\Lambda_X = I_q$; $\longrightarrow$ The measurement error model:

$$\eta = B\eta + \Gamma\xi + \zeta$$

$$Y = \eta + \epsilon$$

$$X = \xi + \delta$$

C) Factor Analysis Models: Just take the measurement part $X = \Lambda_X\xi + \delta$.

IV. Relationship between exploratory and confirmatory FA

In EFA the number of latent variables is not determined in advance; further, the measurement errors are assumed to be uncorrelated. In CFA a model is constructed to a

great extent **in advance**, the number of latent variables ξ is set by the analyst, whether a latent variable influences an observed variable is specified, some direct effects of latent on observed values are fixed to zero or some other constant (e.g. one), measurement errors δ may correlate, the covariance of latent variables can be either estimated or set to any value. In practice, distinction between EFA and CFA is more blurred. For instance, researchers using traditional EFA procedures may restrict their analysis to a group of indicators that they believe are influenced by one factor. Or, researchers with poorly fitting models in CFA often modify their model in an explanatory way with the goal of improving fit.

V. Software available for fitting Structural Equation Models. In the latest version of SAS, the standard procedure **CALIS** for fitting Structural Equation Models has been significantly upgraded in SAS 9.3. In particular, now you can analyse **means and covariance (or even higher order)** structures (instead of just covariance structures like in the classical SEM). Note that the general form of the SEM model given here is only one possible description due to Karl Jöreskog. His paradigm has been first implemented in the software called LISREL (**L**inear **S**tructural **R**elationships).

There are other equivalent descriptions due to Bentler and Weeks, to McDonald and some other prominent researchers in the field. Some of them also have proposed their own software for fitting SEM models according to their model specification. The **EQS** program for PC that deals with the Bentler/Weeks model, was very popular for a while. The latest “hit” in the area is the program MPLUS (M is for Bength Muthén). Mutén is a former PhD student of Jöreskog and has been the developer of LISREL. During the last 15 years or so however, he has developed his own program MPLUS. Its latest version 6 represents a fully integrated framework and is the premier software in the area of general latent variable modelling specifically in the behavioural sciences. Mplus capabilities include:

- Exploratory factor analysis
- Structural equation modeling
- Item response theory analysis
- Growth curve modeling
- Mixture modeling (latent class analysis)
- Longitudinal mixture modeling (hidden Markov, latent transition analysis, latent class growth analysis, growth mixture analysis)
- Survival analysis (continuous- and discrete-time)
- Multilevel analysis
- Bayesian analysis
- etc.